

1. Project ID: **ITP_IT_19**
2. Project Title: Hotel booking & management system
3. Campus: Malabe
4. WD/WE: WD
5. Group Information:
Stream: Information Technology

	Registration Number	Student Name	Phone Number	Signature
01	IT24103110	Wijesinghe W.P.H	070 169 3337	
02	IT23320482	Rashmindika N.H.A.R	076 167 4699	
03	IT24102523	Bambaragama H.M.G.K	076 793 0317	
04	IT24100648	Mandinu W.A.C	077 863 8854	
05	IT24100688	Perera M.R.R	078 690 3375	
06	IT24101162	Perera G.A.R	077 904 3642	

6. Client Information / Project Justification

B. Projects without a Client:

The Hotel Management System was selected because it addresses real-world challenges in managing hotel reservations, room availability, and customer communication. The project allows the implementation of automated features such as email confirmations, booking management, and reporting while applying core concepts of software engineering, databases, and web development.

7. A brief description of the problem.

Many hotels, especially small and medium-scale establishments, rely on manual or semi-automated methods to manage room reservations, availability, payments, and customer communication. These methods often lead to booking errors, double reservations, delayed confirmations, and inefficient record handling. The lack of automation and centralized management makes it difficult for hotels to provide timely services and for customers to access accurate, real-time booking information.

8. Main Features of the Proposed System.

	Registration Number (Same order as above)	Name of Feature/s	Brief description of feature(s) in point form
01	IT24103110	Event & Hall Management	Allow customers to view available event halls and their details (capacity, facilities, pricing). Enable customers to book halls for events such as weddings, conferences, meetings, and parties. Allow selection of event date, time duration, and number of guests. Provide options to request additional services (decoration, catering, audio/visual equipment). Automatically check hall availability to prevent double bookings. Generate booking confirmation and event details. Allow customers to modify or cancel their event bookings.

			Manage hall details (add, update, remove halls). Allow administrators to approve or reject booking requests.
02	IT24102523	Guest Transport Management	Allows guests with confirmed hotel bookings to request arrival pickup and departure drop-off transport Captures pickup or drop-off location, date, and time provided by the guest Links transport requests with the guest's hotel booking details Automatically confirms transport requests based on availability Automatically updates transport status after completion or cancellation Sends automated email notifications for transport confirmations and updates Enables administrators to manage transport services, pricing, and requests

03	IT24100688	Operational Analytics Center (AI)	Uses Generative AI for a conversational interface to query revenue and booking data. Employs predictive models to forecast occupancy and revenue trends for the next 90 days. Features AI-driven anomaly detection to instantly flag unusual system or payment activity. Generates automated daily/monthly performance reports and executive summaries.
04	IT23320482	Payment, Billing & Invoice	Calculates total booking cost including taxes and service charges Processes payments using a mock/sandbox payment gateway Tracks payment status for each booking Automatically generates invoices after successful payment Sends invoices to users via email and handles refunds when applicable

05	IT24100648	Intelligent Guest Relations (AI Chatbot)	Provides 24/7 automated support using NLP for FAQs and policy inquiries. Integrates with the booking engine to allow guests to check status or modify reservations. Features Smart Escalation logic to route urgent guest issues to administrators. Supports multi-language translation for international guests to remove communication barriers.
06	IT24101162	Hotel Room Management with Dynamic Pricing	Allows users to search hotel Room by location and dates Enables hotel staff/admin to add and manage hotels and room types Maintains room availability and room details Applies dynamic pricing based on dates (weekends, seasons) Automatically updates room availability after bookings Handles room locking during the booking process Checks real-time room availability Automatically cancels unpaid bookings after a defined time Applies cancellation policies and charges based on system rules